

Independent Samples

The independent samples can be computed over different periods, used for different stocks. The total number of observations from each algorithm can also differ. As stated above, it is extremely important for the analyst to randomly assign trades to the different algorithms, ensure similar trading characteristics (side, size, volatility, market cap) and market conditions over the trading period. Below are two non-parametric tests that can be used to compare algorithms using independent samples. It is best to compare like algorithms in these tests such as arrival price, IS, aggressive-in-the-money, etc. Since the orders are not split across the algorithms, they can be dynamic and will not compete with one another.

Mann-Whitney U Test

The Mann-Whitney U test compares whether there is any difference in performance from two different algorithms. It is best to compare “like” algorithms in this case (e.g., IS to IS, ultra-aggressive to ultra-aggressive, etc.). The arrival cost metric is the performance metric in this test.

Hypothesis:

- H_0 : Same performance
 H_1 : Different performance

Calculation Process:

1. Let m represent the number of orders transacted by broker A.
 Let n represent the number of orders transacted by broker B.
 Total number of orders = $m + n$.
2. Combine the samples into one group.
3. Order the combined data group from smallest to largest cost.
 For example, the smallest value receives a rank of 1, the second smallest value receives a rank of 2, ..., the largest value receives a rank of $m + n$.
 Identify each observation with an “A” if the observation was from algorithm A and “B” if it was from algorithm B.
4. The test statistic T is the sum of the ranks for all the observations from algorithm A.

This can be computed using help from an indicator function defined as follows:

$$W_i = \begin{cases} 1 & \text{if the observation was from algorithm A} \\ 0 & \text{if the observation was from algorithm B} \end{cases}$$